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Evolutionary game analysis of healthcare data governance in China's experimental data trading pilots

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Han Yao & Yuanli Liu

This study constructs a tripartite evolutionary game model involving healthcare data management authorities (DMAs), healthcare data operating departments (DODs), and data-related entities (DEs) within a triple principal-agent framework. We analyze dynamic interactions among these stakeholders in healthcare data governance, focusing on privacy security, moral hazard, and interest alignment. Results indicate that strategic instability arises under conditions of ambiguous data property rights and asymmetric risk responsibilities. However, the system converges to compliance-oriented equilibria when critical thresholds are surpassed. Notably, DMAs' strong incentive strategies are pivotal in resolving regulatory paradoxes, while DODs' risk behaviors exhibit nonlinear sensitivity to penalty intensity and revenue levels. We further demonstrate that a dual "penalty-compensation" mechanism mitigates economic losses from data breaches, and synergistic government reputation mechanisms with financial incentives reduce regulatory costs. Policy implications include a tiered dynamic regulatory system, a revenue-risk linked distribution mechanism, and a collaborative governance ecosystem driven by technology and credibility constraints.

In the context of the new round of technological revolution and industrial transformation, the explosive growth of data volume and computing power has propelled the digital economy, which takes data as a factor of production and digital technology as the core driving force, to deeply integrate with traditional industries. This integration plays an increasingly important role in transforming economic production modes, enhancing industrial quality and efficiency, and stimulating business model innovation[1,2]. In 2023, the scale of China's digital economy reached 53.9 trillion yuan, accounting for 42.8% of GDP, making it a major driving force for China's economic development[3]. Simultaneously, the vigorous development of the digital economy has given rise to new business forms and service models in the healthcare field. Guided by the Global Strategy on Digital Health 2020–2025 from the World Health Organization (WHO)[4], nations worldwide are actively formulating strategies to advance digital health infrastructure and healthcare data governance frameworks. The European Union has pioneered regulatory innovation through its European Health Data Space (EHDS) initiative launched in 2022[5], which establishes a compliance-driven legal architecture to unlock the full potential of health data reuse while safeguarding citizen rights. Australia's Digital Health Blueprint and Action Plan 2023–2033 emphasizes cross-sector interoperability, standardized electronic health records (EHRs), and secure e-prescription systems to enable trusted data sharing[6]. In the United States, the 2023-2028 HHS Data Strategy prioritizes healthcare digitization through nationwide EHR adoption, enhanced system interoperability, and strengthened privacy-preserving legal frameworks[7]. These global efforts collectively underscore the critical balance between harnessing data-driven healthcare innovation and maintaining rigorous ethical-legal safeguards. Since the Chinese government first proposed the "Internet Plus" action plan in 2015, emphasizing the continuous improvement of medical and health standards and the creation of a Healthy China, the impact of digital intelligence technologies in the healthcare sector has persisted for several years[8]. In October 2015, China's national drug supply security comprehensive management platform was officially launched. By March 2022, the national unified medical insurance information platform was fully completed, and by November 2023, the national health information platform was also largely established. With the milestones of informatization construction in medical insurance, medical care, and medicine ("three medicals") being achieved one by one, a nationwide health governance network system that encompasses the core data of the "three medicals" has begun to take shape, laying a solid foundation for the digital transformation of the coordinated development and governance of the "three medicals" in the new development stage. As a

School of Health Policy and Management, Chinese Academy of Medical Sciences & Peking Union Medical College, Beijing, China.
e-mail: liuyuanli_pumc@163.com

vital foundational strategic resource and industry-specific data element in the health sector, China's vast medical insurance, medical care, and medicine data have generated significant value for people's livelihoods, the economy, and technology[9].

With the continuous expansion of data volume, the supply-demand dynamics of healthcare data have become increasingly diversified and complex. Sole reliance on free data flow and sharing has become inadequate to meet the growing demands for extensive data utilization and protection, necessitating more regulated and effective data circulation mechanisms[10,11]. Data trading, defined as a market-oriented circulation model where providers supply data for compensation and demanders pay acquisition fees using currency as the primary medium of exchange, enables conditional sharing and monetized utilization of data resources[12,13]. To unleash the market potential for public data development and foster high-quality public data product supply, the Communist Party of China (CPC) Central Committee and State Council issued the Opinions on Establishing a Fundamental Data System to Better Leverage the Role of Data Elements (commonly referred to as the "Data Twenty Articles") in 2022. This policy explicitly advocates "exploring conditional paid use of public data for industrial and sectoral development while safeguarding public interests in data supply and utilization"[14]. Under this policy framework, Chinese provincial and municipal governments have successively established big data groups and data exchanges, pioneering global explorations in promoting data resource development through market-oriented mechanisms such as authorized operational transactions[15]. The synergistic effect of central policy making and local implementation has catalyzed the emergence of numerous scenario-specific, value-added public health data products. These innovations include clinical decision support systems enhanced by medical large language models, Helicobacter pylori epidemiological mapping, and AI-driven disease screening models, demonstrating the transformative potential of regulated data markets in healthcare advancement.

However, global research and practices regarding the management and utilization of healthcare data assets remain in their nascent stages. Operational institutions are still exploring and experimenting with various aspects of medical data, including access[16], transmission[17], governance[18], privacy[19], as well as benefits and risks[20]. Beyond technical interoperability, the effective circulation and marketization of healthcare data hinge on enabling multi-agent collaboration and, crucially, on fostering the willingness among stakeholders—such as hospitals, data operators, and regulatory bodies—to engage in sustained cooperation.

Although data integration is essential to tailoring care to patient needs and improving healthcare service quality, multiple structural and behavioral barriers persist. These include capability mismatches between medical institutions and data operators[21], unequal demand structures[22], and mutual distrust among stakeholders[23]. Consequently, the current healthcare market exhibits low willingness and limited extent of data resource sharing among participants, resulting in prominent "data silos" that hinder the realization of scale economies and synergistic effects from vast healthcare data resources[24,25].

While prior studies have primarily focused on the technical challenges of data sharing or the role of digital technologies in advancing the health industry, few have empirically investigated the dynamics of stakeholders' behavioral incentives in data exchange. This study aims to address three pressing and underexplored questions:

(1) How does the circulation of healthcare data reshape strategic decisions and benefit allocation among key stakeholders?
(2) What policy instruments—such as penalty mechanisms or revenue-sharing schemes—can effectively align individual incentives with broader public health goals?
(3) As China becomes the first country to define data as an independent factor of production[26], how can its institutional innovations in data rights, sharing, and governance offer transferable lessons for global digital health policymaking?

To address these questions, we construct a tripartite evolutionary game model involving regulators, data-operating enterprises, and data-owning institutions. This model is parameterized using empirical and policy-relevant data from China's evolving healthcare data factor market. By simulating the impact of different governance tools on stakeholder behavior, the model offers dynamic insights into the policy conditions that can foster sustainable, secure, and equitable data sharing. We contend that these principles, concerning how policy shapes stakeholder behavior towards cooperation, provide a highly transferable framework for optimizing data governance and unlocking the value of healthcare data assets not only in China but also in diverse global contexts facing similar multi-agent coordination challenges.

Results

Analysis of triple principal-agent relationships

In the healthcare data ecosystem, a series of key stakeholders—including healthcare data management authorities (DMAs), healthcare data operation departments (DODs), and healthcare data-related entities (DEs)—participate in strategic decision-making aimed at optimizing their respective utilities. This study reconstructs the analytical framework for healthcare data governance from the perspective of triple principal-agent relationship, as illustrated in Fig. 1. Healthcare data primarily comprises information resources generated during clinical processes. While DEs voluntarily disclose basic personal information during medical consultations, the majority of clinical data – including diagnostic records and treatment parameters – is systematically captured and perpetually utilized by DODs (i.e., healthcare institutions) throughout care delivery cycles. This operational reality fundamentally challenges the applicability of traditional informed consent frameworks in healthcare data ecosystems. Given their triple role as both data custodians and value creators, healthcare institutions inherently assume stewardship over healthcare data assets. To optimize data utility, DODs typically adopt two

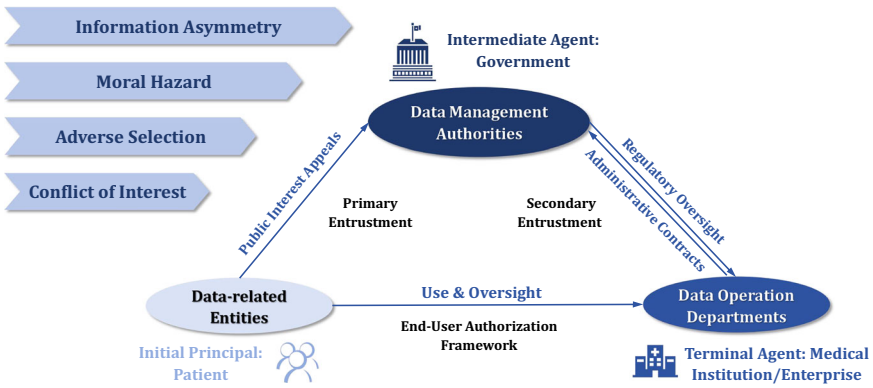

Fig. 1 | Triple principal-agent relationships in healthcare data governance. This schematic depicts the hierarchical governance framework among DMAs, DODs, and DEs, formalizing accountability flows and stakeholder synergies across regulatory, operational, and end-user tiers.

strategic approaches: functioning as independent data processors or forming synergistic alliances with complementary DODs (e.g., insurers and AI developers) through structured data-sharing agreements. These partnerships enable cross-institutional data interoperability, simultaneously driving economic value generation and therapeutic innovation. Healthcare data governance fundamentally constitutes a tripartite institutional mechanism involving data authorization, operational protocols, and management paradigms. At its core, this system facilitates collaborative public value creation among three key stakeholders: DMAs, DODs, and DEs. Based on the governance logic of public goods provision, a hierarchical governance architecture is constructed.

Initial entrustment: public-government fiduciary relationship

DEs constitute the foundational data sources in the healthcare ecosystem, comprising both patients' basic information and secondary data generated through diagnostic and treatment records at healthcare institutions. Healthcare data exhibits hybrid economic propertie: (1) Excludable private good characteristics during primary clinical use; (2) Non-rivalrous public good attributes when shared for research/population health. In the realm of public health services, the quasi-public goods nature of healthcare data renders market mechanisms ineffective for its allocation, leading to a unique principal-agent relationship between the public and the government. While patients may distrust systemic data sharing, they paradoxically demand stronger regulatory safeguards to mitigate misuse risks—a duality addressed through DMA's fiduciary role. As the initial principal, the public entrusts data governance authority to the government, a relationship grounded in three dimensions: First, based on public trust theory, the DMAs bear the constitutional obligation to safeguard citizens' health rights[27]. Second, the sensitivity of healthcare data necessitates public authority intervention for risk control[28]. Third, economies of scale make government-led centralized governance more efficient. This vertical entrustment grants the government legitimate authority over data governance while imposing fiduciary responsibilities to protect public interests and prevent data misuse.

Secondary entrustment: government-operating institution administrative contract relationship

In governance practice, the government delegates data operation rights to specialized DODs through administrative contracts, forming a critical nexus in the governance system[29]. DODs operate through two institutional forms: (1) Healthcare organizations fulfilling dual roles as data custodians and value creators; (2) Licensed third-party entities specializing in data analytics and product development. This layer of principal-agent relationship exhibits three key characteristics: First, explicit contractual relationships are established through access permits and performance contracts. Second, DODs must simultaneously meet public value objectives and market efficiency requirements. Third, DMAs retains regulatory authority to address moral hazards arising from information asymmetry. To address imperfect information, DMAs can employ institutional audits, stakeholder whistleblowing, and blockchain/AI-driven monitoring systems to mitigate asymmetry. Notably, this entrustment is not a simple transfer of power but a complex institutional arrangement encompassing data rights confirmation, benefit distribution, and risk sharing.

End-user authorization framework: operator-public data stewardship contracts

At the level of market-oriented allocation of data factors, a unique market contract relationship forms between DODs and the public. This relationship is characterized by dual constraints: On one hand, DODs obtain data usage rights through service agreements and are obligated to fulfill contractual duties such as privacy protection and quality assurance. On the other hand, the public, as DEs, retains legal rights such as informed consent and benefit distribution. In this process, data assetization and public service provision form a value loop, necessitating effective incentive-compatible mechanisms to balance public interests and commercial value. Importantly, this contractual relationship is nested within the aforementioned institutional frameworks, creating a hybrid governance structure of "administrative regulation + market adjustment".

Table 1 | Parameters and meaning of game behavior

Nomenclature	
Variables	
x	The proportion of DMAs adopting strong incentives strategies.
y	The proportion of DODs that comply with regulations.
z	The proportion of DEs authorizing data.
Parameters	
DMA(s)	
R_1	The basic public welfare revenues brought by strong government incentives.
R_2	The basic public welfare revenues brought by weak government incentives.
F_1	Financial subsidies provided by DMAs under strong incentive policies.
F_2	Financial subsidies provided by DMAs under weak incentive policies.
δ	The policy-induced market expansion multiplier. Each incentive unit increases DODs by $\delta_1 m$ and DEs by $\delta_2 n$. Set $\delta_1 = \delta_2 = \delta$.
I_1	The negative impact on government image due to moral hazard behavior when DODs seek private gains.
P	Penalties for operating departments that exhibit moral hazard.
DOD(s)	
m	Empirically observed number of DODs that exercise data processing rights.[a]
C_{high}	High-quality unit data governance cost.
C_{low}	Low-quality unit data governance cost.
S_1	Rewards for DODs that actively fulfill data governance obligations under strong incentives.
S_2	Rewards for DODs under weak incentives.
K	Data revenues obtained by DODs through the exercise of operational rights.
D	Economic losses due to data breaches.
C	Additional operational costs incurred by DODs when DEs choose not to authorize their data.
DE(s)	
n	Empirically observed number of DEs authorized for data circulation.[b]
V_1	Incentives for individuals who actively authorize data sharing under strong incentives.
V_2	Incentives for individuals authorizing data sharing under weak incentives.
I_2	The trust level of patients in authorized data.
λ	Externality factor. Effective utilization of data.
ΔU	Additional benefits brought by the positive externalities of data sharing.
H	Health benefits obtained by the authorizer from the use of data.
Acronyms	
DMA(s)	Data Management Authority(ies)
DOD(s)	Data Operation Department(s)
DE(s)	Data-related Entity(ies)

[a]Used for baseline calibration or historical benchmarking.
[b]Serves as a historical reference for understanding potential data supply.

Model assumptions

Based on the framework of incomplete information dynamic game theory, this study constructs a tripartite evolutionary game model for healthcare data governance[30,31], proposing the following core hypotheses (Table 1):

Hypothesis 1. The three parties—healthcare data management authorities, data operating departments, and data-related entities—are all under conditions of incomplete information, and each party seeks to maximize its own interests under bounded rationality.

Hypothesis 2. DMAs may actively establish cooperative relationships with relevant institutions, providing comprehensive policy and informational support to protect public interests and patient rights, ensuring the smooth implementation of data-authorized operations. Alternatively, due to considerations of regulatory resources and challenges, as well as the impact of adverse selection caused by information asymmetry, the government may adopt a more passive incentive strategy[32,33]. Therefore, the strategy set for DMAs is *{strong incentives, weak incentives} (SI, WI)*. At the initial time, the probability of DMAs adopting strong incentives is denoted by $x\,(0 \le x \le 1)$, and the probability of weak incentives is $1 - x$.

Hypothesis 3. DODs may choose to comply with government-established data management norms to provide high-quality services or exploit information asymmetry to maximize their own interests, even engaging in moral hazard behaviors that compromise data quality and patient rights[34]. Therefore, the strategy set for DODs is *{following norms, seeking benefits} (FN, SB)*. At the initial time, the probability that DODs follow norms is denoted by $y\,(0 \le y \le 1)$, and the probability of engaging in moral hazard behaviors (seeking benefits) is $1 - y$.

Hypothesis 4. When authorizing DODs to collect, use, and manage personal health data, DEs may choose to trust and authorize or refuse authorization due to concerns about privacy and security. Therefore, the strategy set for DEs is *{authorizing, not authorizing} (A, NA)*. The probability that a patient authorizes data is denoted by $z\,(0 \le z \le 1)$, and the probability of not authorizing is $1 - z$.

Tripartite evolutionary game payoff matrix

Based on the aforementioned assumptions, a tripartite game model involving DMAs, DODs, and DEs is constructed. The variables are defined as follows, with the multi-agent game tree established according to the hypotheses (Fig. 2).

As the initial principal, the public delegates authority to DMAs to oversee the allocation and protection of quasi-public health data. This vertical entrustment grants DMAs both legitimacy and responsibility to design incentive-compatible policies that promote public welfare while deterring moral hazard. To reflect this, we model the DMA's objective function around public welfare enhancement, where the baseline social revenues under different incentive regimes are represented by R_1 (strong incentives) and R_2 (weak incentives). Under weak incentives, a baseline number of m DODs implement data processing usage rights and n DEs authorize data circulation, where $m \in [0, +\infty)$, $n \in [0, N]$. These values (m and n) are empirically observed quantities derived from historical or real-world data, and serve as reference parameters for calibrating initial market conditions[35]. If the DMA adopts strong incentives, it can attract more DODs and DEs to enter the market, thereby generating economies of scale. The incentive factor δ is introduced to represent its positive impact on the market, leading to the addition of $\delta_1 m$ DODs that implement data processing usage rights and $\delta_2 n$ DEs who authorize data circulation. To characterize the scale of interactions between DODs and DEs, the multiplicative term $m \times n$ is employed. This form captures the bilateral activation effect: meaningful data-sharing outcomes emerge only when both sides actively participate. The interaction scale grows with potential cross-entity linkages rather than being constrained by the smaller group size. The DMA will reward DODs that actively fulfill data governance duties and comply with norms with an incentive of S_1, while also incentivizing DEs who actively authorize data sharing with V_1. Under weak incentives, the reward for DODs is reduced to S_2, and the reward for DEs who authorize data sharing is reduced to V_2. When DODs seek private gains during this process, the DMA's work becomes more challenging, and the resulting moral hazard behaviors lead to a loss of DMA image, which we define as I_1. At this point, the DMA will impose a penalty P on the DODs engaging in moral hazard, with the penalty considered equivalent to the effect of taxation.

DODs are delegated operators (either healthcare institutions or certified third parties) that process data under formal contracts with DMAs. This secondary entrustment forms a horizontal administrative relationship grounded in access control, performance obligations, and public value delivery. DODs face dual pressures: regulatory compliance and commercial viability. These m DODs, through active data governance, cannot only enhance the value of data resources but also gain economic benefits from them. The data governance process involves the collection, storage, processing, and privacy protection of data resources, resulting in governance costs for high-quality data (C_{high}). Some DODs, in an effort to save on governance costs or pursue higher returns, may fail to comply with data governance standards, divert data resources to informal channels, and adopt low-quality data, thus incurring lower governance costs (C_{low}). Research on data quality management by IBM shows that the governance costs of low-quality data are, in fact, higher than those of high-quality data[36]. Specifically, in areas such as data cleansing, deduplication, and format standardization, low-quality data typically requires more resources[37]. Under a strong incentive policy environment, DODs can receive financial subsidies F_1 from the government for active data governance, whereas under a weak incentive policy, this subsidy reduces to F_2. DODs can also obtain data revenue K through exercising business rights, but due to inherent system risks and incomplete regulation, data breaches may occur under all strategy combinations, resulting in an economic loss D.

Patients, as DEs, have the decision-making power to choose whether to authorize the use of their personal health data. In the context of compliant operations by DODs that generate data value, n DEs authorize data circulation. Simultaneously, due to the existence of psychological accounts, patients develop trust in their authorized data (I_2). When patients choose to

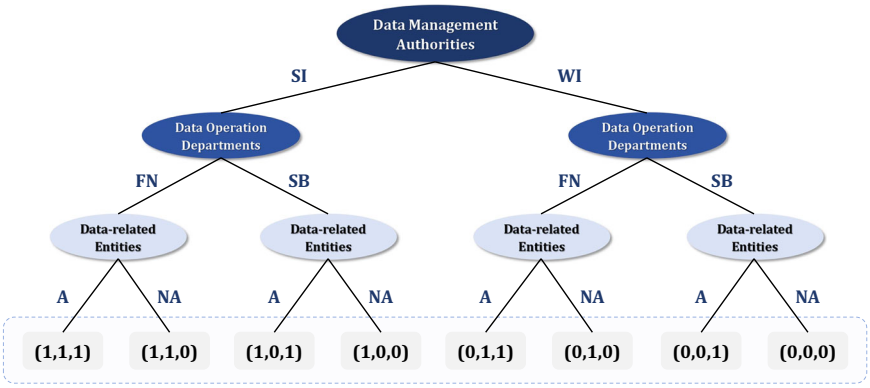

Fig. 2 | Multi-agent game tree diagram. The game tree initiates with DMAs selecting regulatory strategies {Strong Incentives (SI) or Weak Incentives (WI)}, followed by DODs choosing operational approaches {Following Norms (FN) or Seeking Benefits (SB)}, and terminating with DEs determining data-sharing consent {Authorizing (A) or Not Authorizing (NA)}. Terminal nodes depict equilibrium outcomes as ternary tuples *(x,y,z)*, where: x = probability of DMA's strong incentives (SI), y = probability of DOD's norm compliance (FN), z = probability of DE's data authorization (A).

Table 2 | Tripartite evolutionary game payoff matrix

			DODs		
				Following norms (y)	Seeking benefits (1-y)
DMAs	strong incentives (x)	**DEs**	authorizing (z)	$R_1 - S_1 - V_1 - F_1$ $S_1 + (1+\delta_1)(1+\delta_2)mn(K - C_{high} - D) + F_1$ $V_1 + (1+\delta_2)nI_2 + \frac{(1+\delta_1)(1+\delta_2)mn}{N}H + \lambda\Delta U$	$R_1 - V_1 + (1+\delta_2)n(P - I_1)$ $(1+\delta_2)mn(K - C_{low} - D) - (1+\delta_2)nP$ $V_1 + \frac{(1+\delta_2)mn}{N}H + \Delta U$
			not authorizing (1-z)	$R_1 - S_1 - F_1$ $S_1 + (1+\delta_1)mn(K - C_{high} - D) + F_1 - C$ $nI_2 + \frac{(1+\delta_1)mn}{N}H$	$R_1 + n(P - I_1)$ $mn(K - C_{low} - D) - nP - C$ $\frac{mn}{N}H$
	weak incentives (1-x)	**DEs**	authorizing (z)	$R_2 - S_2 - V_2 - F_2$ $S_2 + mn(K - C_{high} - D) + F_2$ $V_2 + nI_2 + \frac{mn}{N}H + \lambda\Delta U$	$R_2 - V_2 + n(P - I_1)$ $mn(K - C_{low} - D) - nP$ $V_2 + \frac{mn}{N}H + \Delta U$
			not authorizing (1-z)	$R_2 - S_2 - F_2$ $S_2 + mn(K - C_{high} - D) + F_2 - C$ $nI_2 + \frac{mn}{N}H$	$R_2 + n(P - I_1)$ $mn(K - C_{low} - D) - nP - C$ $\frac{mn}{N}H$

NOTE. To reflect the bilateral activation effect between DODs and DEs, the interaction scale is modeled as *mn*, representing the total potential of cross-entity engagement rather than one-to-one matching.

authorize their data, it provides the foundation for the circulation and utilization of healthcare data, benefiting medical institutions and enterprises in clinical research, disease monitoring, and the provision of personalized medical services[38]. The framework introduces a positive externality coefficient (λ) to quantify the effective utilization rate of healthcare data. This coefficient exhibits positive correlation with the intensity of data sharing externalities, which manifest through three distinctive characteristics: high liquidity, low marginal replication costs, and inherent increasing returns to scale[39,40]. As λ increases, the shared data generates augmented benefits (ΔU) distributed across three primary beneficiaries: healthcare providers (enhanced diagnostic capabilities), researchers (improved analytical validity), and patients (optimized therapeutic outcomes). Concurrently, DEs obtain health utility gains (H) derived from secondary data applications. The decision-making efficacy within healthcare systems demonstrates proportional enhancement relative to the data acquisition scale of DODs, following the principle of n/N representativeness[41]. However, DEs may also choose not to authorize their data due to concerns over privacy and security. These concerns may stem from the risks of data breaches, misuse, or abuse[42]. When DEs choose not to authorize their data, the difficulty for DODs in acquiring data increases, potentially limiting data-driven medical innovations and service improvements, and imposing additional operational costs (C) on the operating departments.

The payoff matrix for the tripartite evolutionary game involving DMAs, DODs, and DEs is shown in Table 2.

Stability analysis of system equilibrium points

Since mixed strategy equilibria in asymmetric dynamic games are definitely not evolutionarily stable equilibria, only the pure strategy equilibrium points of the evolutionary game system are analyzed. Let $F(x) = 0, F(y) = 0, F(z) = 0$, the evolutionary game system has 8 pure strategy equilibrium points, respectively $E_1(0,0,0)$, $E_2(1,0,0)$, $E_3(0,1,0)$, $E_4(0,0,1)$, $E_5(1,1,0)$, $E_6(1,0,1)$, $E_7(0,1,1)$ and $E_8(1,1,1)$. To analyze the stability of the equilibrium points, the Lyapunov first method (indirect method) is used for determination. First, construct the Jacobian matrix (Supplementary materials):

$$J = \begin{bmatrix} \frac{\partial F(x)}{\partial x} & \frac{\partial F(x)}{\partial y} & \frac{\partial F(x)}{\partial z} \\ \frac{\partial F(y)}{\partial x} & \frac{\partial F(y)}{\partial y} & \frac{\partial F(y)}{\partial z} \\ \frac{\partial F(z)}{\partial x} & \frac{\partial F(z)}{\partial y} & \frac{\partial F(z)}{\partial z} \end{bmatrix} \quad (1)$$

According to the Lyapunov first method (indirect method), if all three eigenvalues are negative, then the equilibrium point is an evolutionarily stable point; if all three eigenvalues are positive, then the point is an unstable point; if one or two are positive, then the point is a saddle point. The stability of each equilibrium point is shown in Table 3.

For the sake of discussion, let $B = F_1 - C + S_1 + \delta mnC_{high} + (1+\delta)mnC_{low} + n\delta P$, and assume that the data revenue K obtained by the DODs from exercising operational rights is low when $K < D - \frac{B}{(2\delta+1)mn}$, and high when $K > D - \frac{B}{(2\delta+1)mn}$, during which the revenue of the DODs increases with the increase in economic loss risk.

Corollary 1. The equilibrium point $E_4(0,0,1)$ requires the following conditions for a stable strategy: $S_2 + F_2 < mn(C_{low} - C_{high})$ and $R_1 < R_2 + n(I_1 - P)$. In other words, when DMAs provides weak incentives and DEs opt to grant data rights, if the financial subsidies and rewards received by DODs are less than the additional data governance costs incurred by using low-quality data, the DODs will still choose to engage in morally risky behavior. Therefore, to prevent the emergence of the $E_4(0,0,1)$ strategic equilibrium, the DMAs should increase the penalties P for morally risky behavior by the DODs, as well as the financial subsidies F_2 and rewards S_2 under compliant operational scenarios, to avoid the realization of these conditions.

Corollary 2. The equilibrium point $E_6(1,0,1)$ requires a stable strategy under the condition: $K < D - \frac{B}{(2\delta+1)mn}$. This means that when the DMAs provides strong incentives and DEs choose to grant data rights, if the data revenue obtained by the DODs from exercising operational rights is less than the economic risk brought by their unauthorized profit-seeking behavior and data leakage, they will still opt for morally risky actions even if they must bear certain economic responsibilities in the data governance process. The motivation behind morally risky actions (such as non-compliance with data protection regulations and misuse of data) stems from the potential for short-term economic benefits. Therefore, to prevent the emergence of the $E_6(1,0,1)$ strategic equilibrium, DMAs should increase the penalties P for morally risky behavior by DODs, enhance the revenue K obtained by the data operation, and improve the financial subsidies F_1 and rewards S_1 under compliant operational scenarios, thus avoiding the realization of these conditions.

Corollary 3. The equilibrium point $E_7(0,1,1)$ requires a stable strategy under the conditions: $S_2 + F_2 > mn(C_{high} - C_{low})$ and $S_2 + F_2 > R_1 - R_2 + F_1 + S_1 + (1-\delta)n(I_1 - P)$. These conditions demand that under the weak incentives from the DMAs, the total rewards and subsidies from the DODs must exceed the cost difference between high-quality and low-quality data governance. This ensures that even with weaker incentives, the DODs have sufficient economic support to engage in high-quality data governance rather than opting for low-quality measures to save costs. Additionally, the total rewards and subsidies must exceed the difference between the financial subsidies and rewards under strong incentives and the penalties after considering the impact of morally risky behavior. This determines that the DODs will not engage in morally risky actions even

Table 3 | The evolutionary stability and stability conditions of each equilibrium point

Points	Eigenvalues λi	Symbol	Stability	Stable conditions
$E_1(0,0,0)$	$\lambda_1 = R_1 - R_2$ $\lambda_2 = \Delta U + V_2$ $\lambda_3 = F_2 + S_2 - mn(C_{high} - C_{low})$	(+, +, ×)	Saddle point	/
$E_2(1,0,0)$	$\lambda_1 = R_2 - R_1$ $\lambda_2 = \Delta U + V_1 + \frac{\delta mn}{N}H$ $\lambda_3 = F_1 + S_1 - mn(C_{high} - C_{low})$ $-\delta mn(D - C_{high} - K)$	(-, +, ×)	Saddle point	/
$E_3(0,1,0)$	$\lambda_1 = \lambda\Delta U + V_2$ $\lambda_2 = mn(C_{high} - C_{low}) - S_2 - F_2$ $\lambda_3 = F_2 - F_1 + R_1 - R_2 - S_1 + S_2$	(+, -, -)	Saddle point	/
$E_4(0,0,1)$	$\lambda_1 = -\Delta U - V_2$ $\lambda_2 = F_2 + S_2 + mn(C_{high} - C_{low})$ $\lambda_3 = R_1 - R_2 - n(I_1 - P)$	(-, ×, ×)	ESS	$F_2 + S_2 < mn(C_{low} - C_{high})$ $R_1 < R_2 + n(I_1 - P)$
$E_5(1,1,0)$	$\lambda_1 = mn(C_{high} - C_{low}) - S_1 - F_1$ $+\delta mn(C_{high} + D - K)$ $\lambda_2 = F_1 - F_2 - R_1 + R_2 + S_1 - S_2$ $\lambda_3 = \lambda\Delta U + V_1 + \delta nI_2 + \frac{\delta(1+\delta)mn}{N}H$	(×, ×, +)	Saddle point	/
$E_6(1,0,1)$	$\lambda_1 = R_2 - R_1 + n(I_1 - P)$ $\lambda_2 = -\Delta U - V_1 - \frac{\delta mn}{N}H$ $\lambda_3 = F_1 - C + S_1 + \delta mnC_{high} + (1+\delta)mnC_{low}$ $-(2\delta+1)mn(D - K) + n\delta P$	(-, -, ×)	ESS	$K < D - \frac{B}{(2\delta+1)mn}$
$E_7(0,1,1)$	$\lambda_1 = -\lambda\Delta U - V_2$ $\lambda_2 = mn(C_{high} - C_{low}) - S_2 - F_2$ $\lambda_3 = F_2 - F_1 + R_1 - R_2 - S_1$ $+S_2 + (\delta - 1)n(I_1 - P)$	(-, -, ×)	ESS	$R_1 < R_2 + F_1 - F_2$ $+S_1 - S_2 + (1-\delta)n(I_1 - P)$
$E_8(1,1,1)$	$\lambda_1 = -\lambda\Delta U - V_1 - \delta nI_2 - \frac{\delta(1+\delta)mn}{N}H$ $\lambda_2 = F_1 - F_2 - R_1 + R_2 + S_1 - S_2 + (1-\delta)n(I_1 - P)$ $\lambda_3 = -F_1 + C - S_1 - \delta mnC_{high} - (1+\delta)mnC_{low}$ $+(2\delta+1)mn(D - K) - n\delta P$	(-, ×, ×)	ESS	$R_1 > R_2 + F_1 - F_2 + S_1 - S_2$ $+(1-\delta)n(I_1 - P)$ $K > D - \frac{B}{(2\delta+1)mn}$

NOTE. $B = F_1 - C + S_1 + \delta mnC_{high} + (1+\delta)mnC_{low} + n\delta P$, × indicates that the positive or negative status of the eigenvalue cannot be determined.

in a weak incentive environment, thereby safeguarding data security and enhancing data governance quality, which is the ideal state for the circulation of healthcare data factors.

Corollary 4. The equilibrium point $E_8(1,1,1)$ requires a stable strategy under the conditions: $K > D - \frac{B}{(2\delta+1)mn}$ and $R_1 > R_2 + F_1 - F_2 + S_1 - S_2 + (1-\delta)n(I_1 - P)$. That is, under strong incentive policies, the total revenue of DMAs and DODs must exceed the total revenue under weak incentive policies plus the potential penalties for considering morally risky behavior. At the same time, when facing data leakage risks, the data revenue K obtained by the DODs through compliant operations can exceed the potential economic losses D minus certain costs and additional revenue $B = F_1 - C + S_1 + \delta mnC_{high} + (1+\delta)mnC_{low} + n\delta P$. This condition is crucial for encouraging responsible data management practices by DODs, as it ensures they receive sufficient economic incentives while adhering to regulations. Through such policy design, a stable equilibrium can be promoted where all participants can achieve their interests while ensuring data security and privacy, thereby facilitating data sharing and circulation and further realizing the positive externalities of data.

Parameter analysis

Impact of initial probabilities on subject strategies. We set $x = y = z = 0.1, 0.2, 0.4, 0.5, 0.7, 0.8$, representing various scenarios where the initial probabilities of strategy choices for each subject range from low to high. The strategic evolution of DMAs, DODs, and DEs is illustrated in Fig. 3. This study assigns parameters based on the current context of healthcare data factor circulation in China, and initial probabilities have varying impacts on all three parties. Under current parameter settings, the lower the initial probability, the faster the strategic evolution of DODs and DEs. Under lower initial probabilities (Fig. 3a–d), the system exhibits oscillations due to weak DMAs incentives, while higher probabilities (Fig. 3e, f) lead to stabilization at $E_4(0,0,1)$. Concurrently, during this period, the health data factor market is still in its infancy, with policies and regulations, privacy protection, and data product development still underway. DMAs have not yet played a guiding role in the market and mainly adopt weak incentive measures. Therefore, the evolutionary stable states $E_7(0,1,1)$ and $E_8(1,1,1)$ are the desired states under current conditions. This study will conduct simulation analysis around this ideal stable strategy. Considering the current level of marketization of healthcare data factors and the assumption of symmetric bounded rationality, the initial strategy selection probabilities are set as: $x = 0.4$, $y = 0.4$, $z = 0.4$.

Impact of moral Hazard penalty *P* on strategic behavior. According to Corollaries 1, 2, 3, and 4, the system's stable points vary depending on the revenue generated by data and the penalties P imposed on DODs for moral hazard behaviors. Therefore, the simulation analysis considers two scenarios: high, medium, and low penalties, as well as low and high revenues.

Assigning P values of *200, 2000, and 3000*, and K values of *3542.86, 5000, 6000, …, 9000*, the replication dynamic equations are iterated 10^4 times, yielding the results shown in Fig. 4. Figure 4a illustrates the system evolution under low penalties. The curves of y and x over time indicate significant oscillations before the system stabilizes, suggesting that low penalties are insufficient to effectively deter moral hazard behaviors, leading to system instability. DODs may adopt opportunistic behaviors due to the low cost of penalties, resulting in significant performance fluctuations. Figure 4b, c depict the system evolution under medium and high penalties, respectively. The curves of y and x over time show that the system stabilizes more quickly, with reduced oscillations. The 3D steady-state diagrams reveal that, compared to the low-penalty scenario, the system reaches stable equilibrium points under medium and high penalties. When penalties are moderate, the system eventually converges to equilibrium points $E_6(1,0,1)$

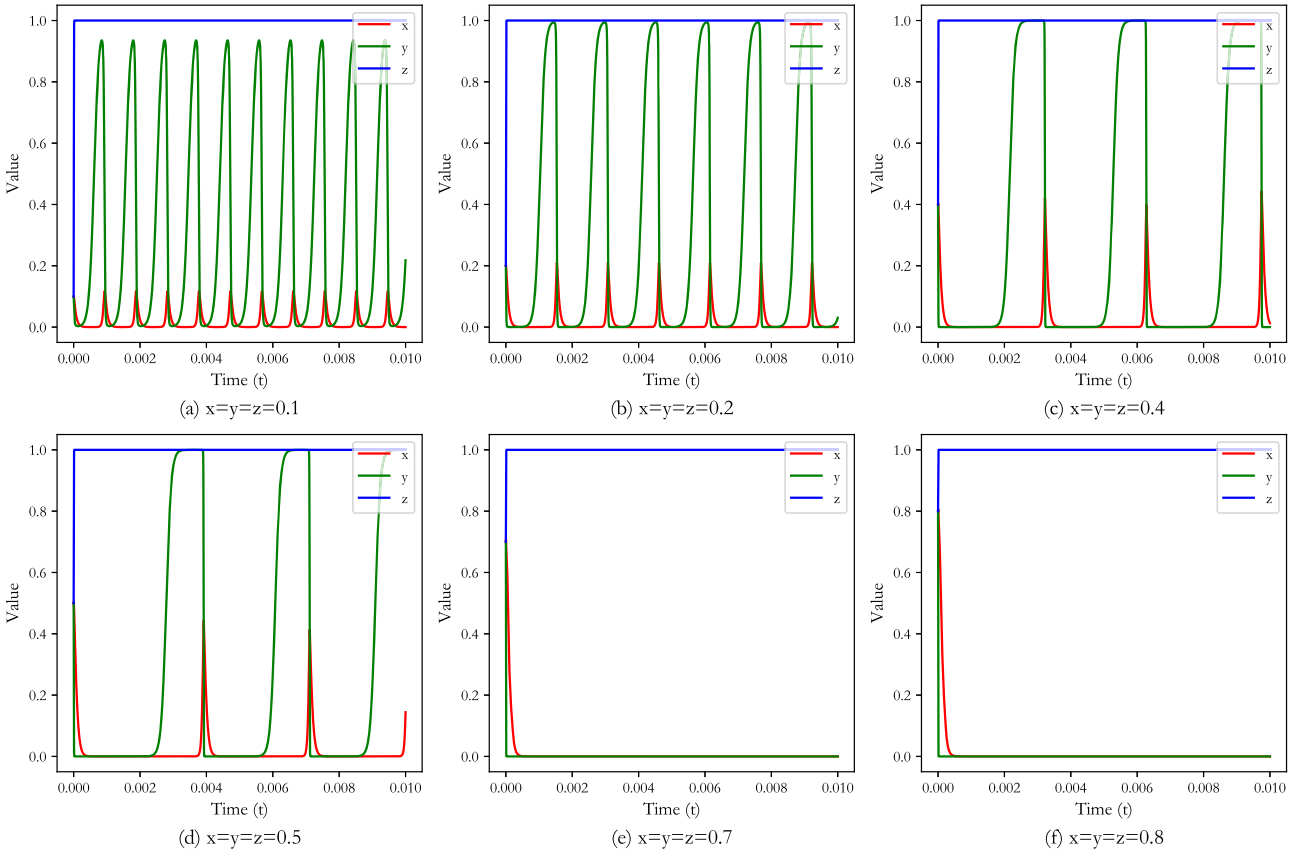

Fig. 3 | Tripartite dynamic evolutionary paths under different initial probabilities. Phase-space trajectories depicting tripartite strategic convergence across six initial probability configurations.

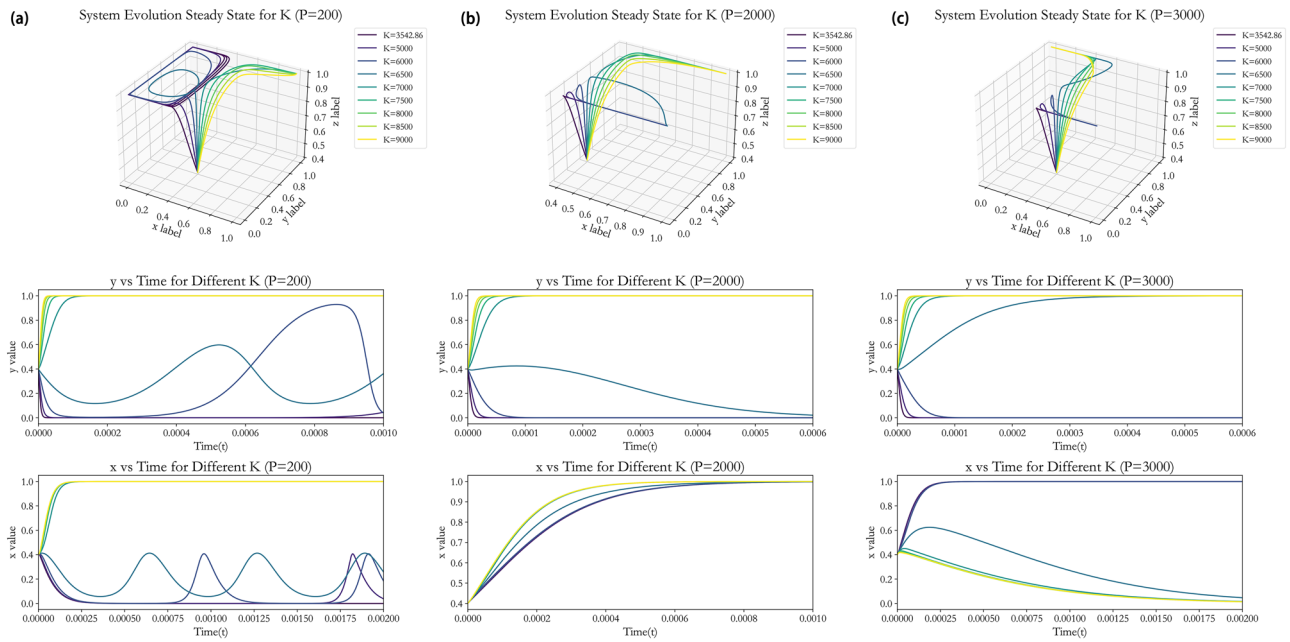

Fig. 4 | Impact of moral hazard penalty *P* and data revenue *K* on system evolution. Three panels contrast governance outcomes under incremental moral hazard penalties: **a** low Penalty; **b** medium Penalty; **c** high penalty.

or $E_8(1,1,1)$. Under high penalties, the system converges to $E_6(1,0,1)$ or $E_7(0,1,1)$.

Impact of data revenue *K* on strategic behavior. As shown in Fig. 4a, under low penalties and low revenue, the system exhibits significant oscillations. Specifically, the system is more likely to reach $E_4(0,0,1)$ rather than $E_8(1,1,1)$, indicating that low revenue is insufficient to incentivize DODs to adopt stable compliance strategies. DODs may participate less in data operations or engage in illegal activities due to low revenues, leading to system instability. Figure 4b shows that under

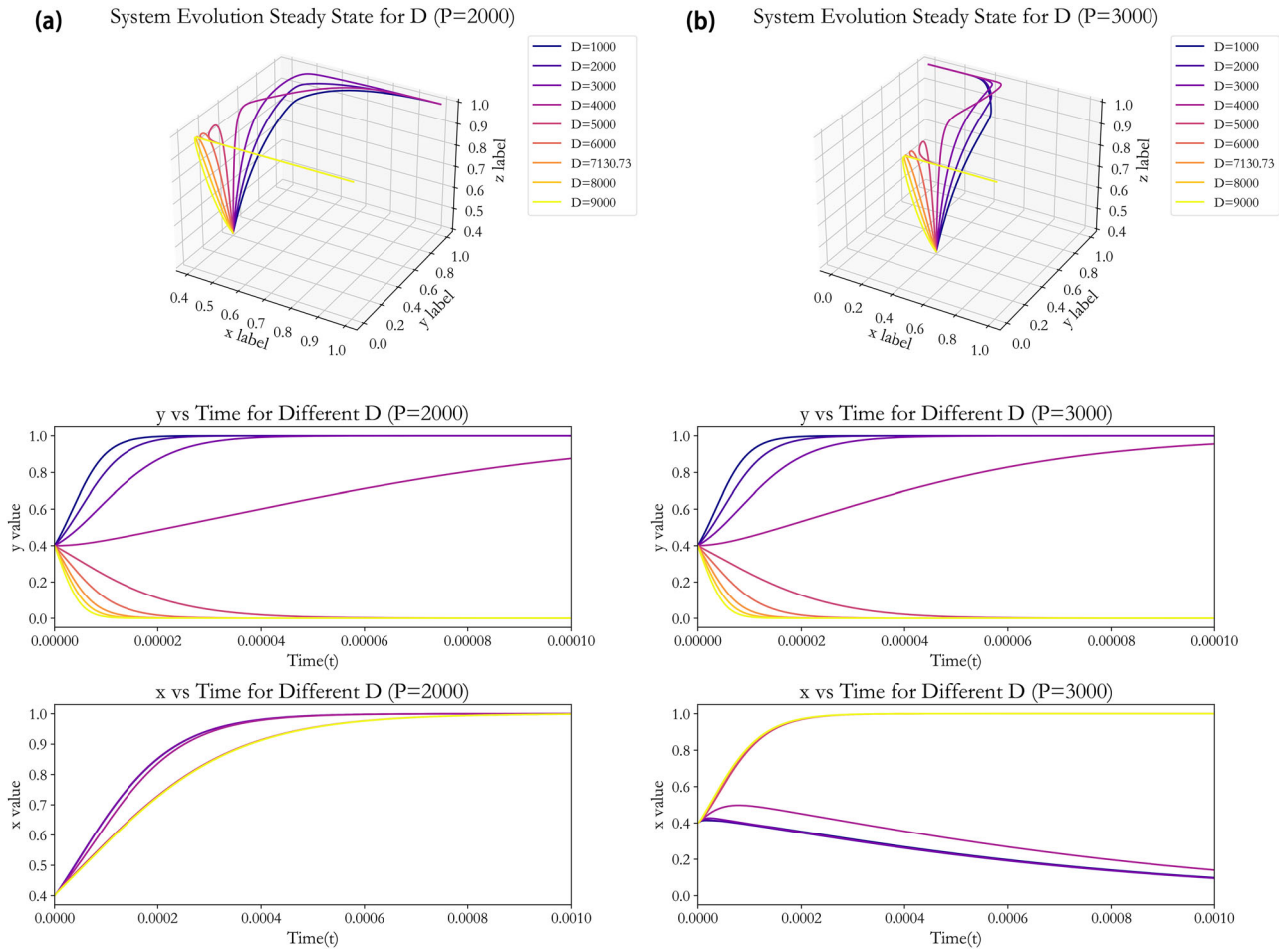

Fig. 5 | Impact of data breach loss *D* on system evolution. Two panels contrast governance outcomes under incremental data breach impacts: **a** Medium Penalty; **b** High Penalty.

medium penalties, the strategy choices of DMAs remain relatively stable and proactive throughout the process. As data revenues increases, DODs shift from non-compliance to compliance, and the system eventually converges to $E_8(1,1,1)$. Figure 4c demonstrates that under high penalties, the strategy choices of both DMAs and DODs are sensitive to data revenue K. As data revenue increases, their strategies gradually become more standardized and cautious. High revenue effectively curbs the risk behaviors of DODs while prompting DMAs to adopt weak incentives, ultimately driving the system toward equilibrium point $E_7(0,1,1)$.

Impact of data breach loss *D* on strategic behavior. Assigning D values of *1000, 2000, 3000, …, 9000*, the replication dynamic equations are iterated 10^4 times, yielding the results shown in Fig. 5. Figure 5a illustrates the system evolution under medium penalties, where DMAs tend to adopt strong incentives. When the economic loss is low, the steady state of the system evolution shows that DODs tend to comply, i.e., the system is more likely to stabilize at $E_8(1,1,1)$. However, as D increases, the system gradually stabilizes at $E_6(1,0,1)$, indicating that under the pressure of high economic losses, DODs tend to adopt risky non-compliant strategies. Figure 5b shows the system evolution under high penalties, where the system is more sensitive to changes in D. When D is low, the system already shows a tendency to stabilize at $E_7(0,1,1)$, but as D further increases, the system stabilizes at $E_6(1,0,1)$. This indicates that under high penalties, even small data breach losses are sufficient to prompt DODs to adopt stricter compliance strategies. When formulating data security management policies, it is necessary to comprehensively consider the impact of moral hazard penalties and data breach losses to guide DODs toward reasonable strategies, thereby improving the stability and efficiency of the entire system.

Impact of fiscal subsidy F_1 and reward S_1 oN Strategic Behavior. To further analyze the impact of DMAs actions on system evolution, this study assigns a series of values to F_1 and S_1 under medium subsidies ($P = 2000$) and relatively high data revenues ($K = 7000$), and simulates the system evolution over time using the replication dynamic equations. Specifically, F_1 ranges from 1500 to 13,000, and S_1 ranges from 100 to 6000. The simulation results are shown in Fig. 6. Figure 6a illustrates the system evolution under different F_1 values. The strategic choices of system agents do not change with increasing F_1, always stabilizing at $E_8(1,1,1)$. As F_1 increases, the system stabilizes more quickly, indicating that higher financial subsidies prompt DODs to adopt compliance strategies more rapidly. Figure 6b shows the system evolution under different S_1 values. As S_1 increases, the system becomes more sensitive to changes in S_1. When S_1 is low, the system already shows a tendency to stabilize at $E_7(0,1,1)$, but as S_1 further increases, the system stabilizes at $E_8(1,1,1)$. Both states are desirable stable states, reflecting the importance of reward mechanism design for policymakers. Appropriately increasing rewards can more effectively guide the behavior of DODs, promoting system stability and healthy development.

Impact of government reputation mechanism I_1 and market incentive factor δ on strategic behavior. The government reputation mechanism I_1 and the market incentive factor δ, as key factors influencing the strategic choices of DMAs and DODs, play important roles in system

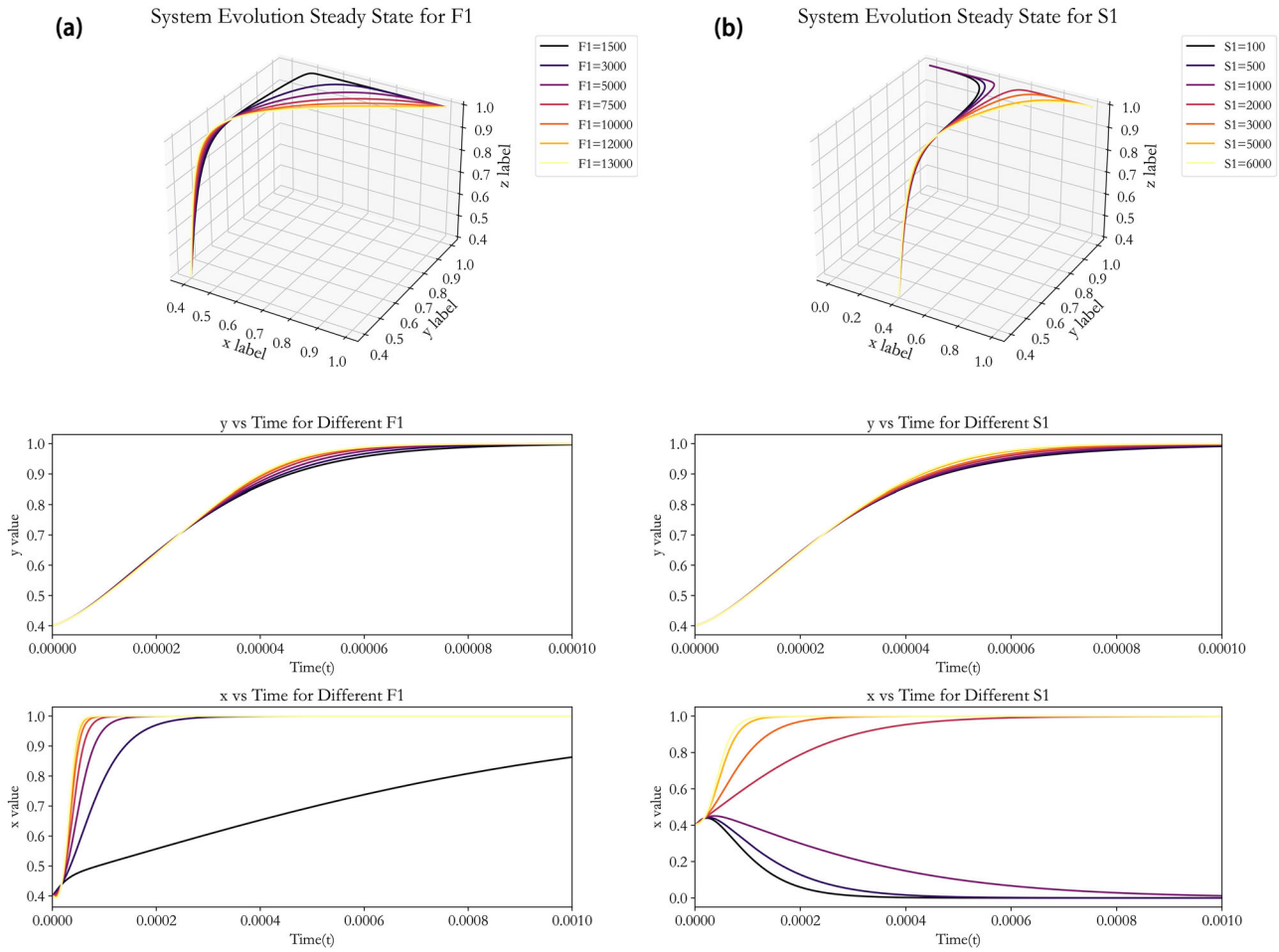

Fig. 6 | Impact of government actions on system evolution. Two panels contrast distinct fiscal intervention mechanisms: **a** Fiscal Subsidy F_1 ; **b** Reward S_1.

evolution. Assigning I_1 values from 100 to 3000 and δ values from 0.5 to 5, their impact on system evolution is analyzed, and the replication dynamic equations are used to simulate the system evolution over time, yielding the results shown in Fig. 7. The DMA reputation mechanism I_1 reflects the loss of government reputation caused by the moral hazard behaviors of DODs. Figure 7a illustrates the system evolution under different I_1 values. The strategic choices of DODs are not affected by I_1. When I_1 is low, the government reputation mechanism has a weak constraining effect on DMAs, which tend to adopt weak incentives to pursue higher revenue, and the system may stabilize at $E_7(0,1,1)$. However, as I_1 increases, the system stabilizes at $E_8(1,1,1)$, indicating that damage to government reputation due to improper behaviors of DODs increases the concerns of DMAs about non-compliance, fearing negative impacts on their reputation. Therefore, DMAs are more likely to adopt strong incentives to maintain good relationships with DODs and DEs and avoid potential reputational losses. The market incentive factor δ reflects the positive impact of strong incentives on the market. As shown in Fig. 7b, regardless of changes in δ, the system eventually stabilizes at a consistent strategy combination. This indicates that under the current system settings, the market incentive factor δ is not a key factor influencing the strategic choices of agents.

Discussion

Advancing the market-oriented allocation of healthcare data factors while effectively protecting patient privacy, corporate rights, and public safety holds significant importance for fostering new economic growth drivers and modernizing national governance. By constructing a tripartite evolutionary game model involving DMAs, DODs, and DEs, this study dynamically analyzes the temporal evolution of strategic interactions while accounting for the preferences and behaviors of agents operating within the triple principal-agent framework. The research thoroughly investigates the cooperative and competitive dynamics among these agents and examines how their strategies evolve with changes in healthcare data governance landscapes. Based on these findings, this paper proposes policy optimization pathways across three dimensions: tiered regulation, data revenue distribution, and incentive refinement.

Studies have found that one of the most trusted data processors by the public for the use of personal information is often clinicians and healthcare organisations[43]. In the patient-healthcare organisation relationship, it is rare for a patient to refuse to be served without providing personal information. Because of this, it is all the more important for DODs to comply with the requirements of laws related to personal information under their legal frameworks, take appropriate protection measures, and handle patient healthcare data securely so that the trust relationship between doctors and patients can be made stronger[44]. In view of the limitations of the existing penalty standard of China's Data Security Law in the field of healthcare, it is proposed to establish a two-tier penalty mechanism of "floating benchmark + progressive behavior". According to Articles 28 to 32 of The Personal Information Protection Law of China, "biometric information" and "medical and health information" are included in the scope of "sensitive personal information"[45]. Drawing on the "4% of global turnover" stepped fine mechanism of the EU General Data Protection Regulation[46], the penalty benchmark is deeply tied to data sensitivity. For example, if core privacy data such as genetic and diagnostic records are involved, the penalty amount will be adjusted to 5-10 times of the general data; and 3-5 times of the penalty will be applied to health behavior data. A progressive enforcement rule of "initial

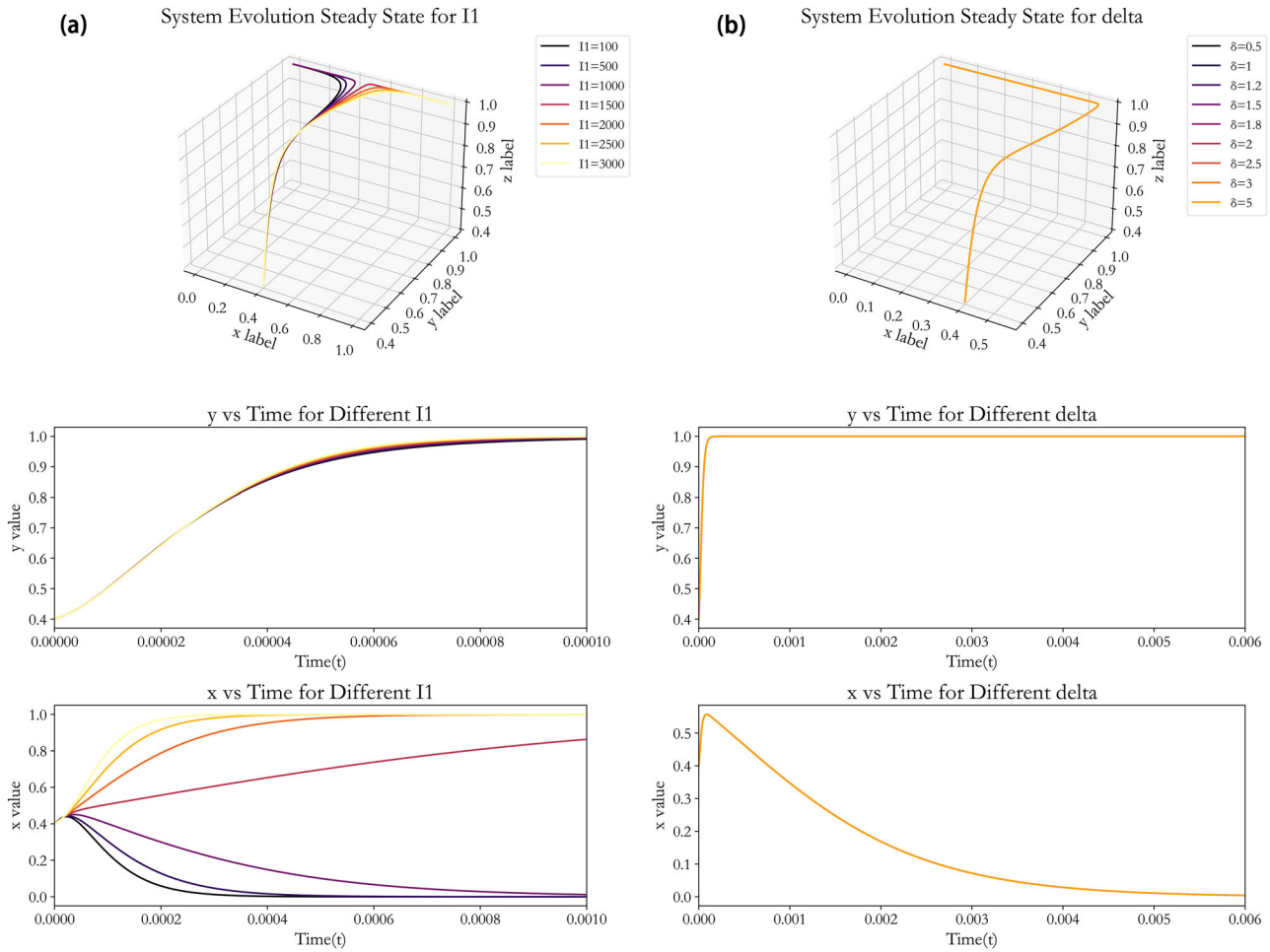

Fig. 7 | Impact of externality parameters on system evolution. Two panels contrast non-monetary governance factors: **a** Government Reputation Mechanism I_1; **b** Market Incentive Factor δ.

warning, second penalty, third suspension" has been established, and a system of multiplying the penalty coefficients has been implemented for organisations that repeatedly violate the law within a year.

In terms of the data revenue distribution system, the plurality of data value creation should be at the core, and fair revenue distribution rules based on specific roles and contributions should be used to distribute economic and social value to DEs in the data value chain. Firstly, the right to shared benefits of the data source, i.e. patients, should be fully guaranteed. As the core contributors of the data, the diagnosis and treatment data provided by patients lay the foundation for medical research and technological innovation, so patients should enjoy the corresponding right to know and the right to benefit from data sharing. Secondly, the right to profit of data holders and operators should also be clarified. DODs have invested a lot of resources in the process of data collection, storage and management, and reasonable revenue distribution is a recognition and incentive for their input. On the basis of legally authorised data use, they are entitled to a corresponding share of revenue. At the same time, data users use healthcare data to research and develop new medicines or diagnostic and treatment technologies, and their innovations bring great economic and public benefits to the society, and they should also be duly rewarded in the revenue distribution[47]. Therefore, it is recommended that a distribution system of "positive incentives + negative constraints" be constructed under the framework of the "Reform Program for Market-based Allocation of Data Factors". Provide policy tilts (e.g., tax breaks) for low-yield and high-public welfare data operation scenarios to make up for compliance costs; and mandatorily withdraw risk reserves for high-yield and high-risk scenarios, with the ratio rising in steps with the growth of revenues. Establish a recovery mechanism linking data leakage losses to revenues, and require DODs to bear liability in proportion to their revenues. For example, 5-15 per cent of net profit can be withdrawn as a risk reserve, to be included in the unified management of the National Health Insurance Fund.

Focus on optimising financial subsidies and reputation mechanisms to form a "technology-driven + credit-constrained" incentive combination. Financial subsidies should focus on supporting compliance technology (e.g. privacy computing) and avoid directly subsidising operational revenues, while the incentive mechanism should focus on long-term compliance behaviors, with a positive correlation function of "compliance duration - incentive coefficient"[48]. Incorporate the reputation mechanism into the regulatory assessment, and implement administrative accountability for reputation loss triggered by departmental dereliction of duty, to force the choice of strong incentive strategies. Build a trustworthy platform for the circulation of data factors, and use smart contracts to automatically enforce the rules of penalties, rewards, and revenue distribution, so as to reduce the space for human intervention. Establishing a compliance credit score system for healthcare data, with the score level linked to market access and policy support, forming a long-term mechanism with compatible incentives. Using the National Healthcare Big Data Center as a carrier, a three-in-one governance ecosystem of "system-technology-market" will be constructed.

This study systematically investigates the governance mechanisms of healthcare data factors through an evolutionary game model embedded in a triple principal-agent framework, offering critical insights into optimizing market-oriented data allocation while balancing privacy protection and public value creation. The findings underscore the pivotal role of calibrated penalty mechanisms in curbing moral hazards, the necessity of benefit-risk

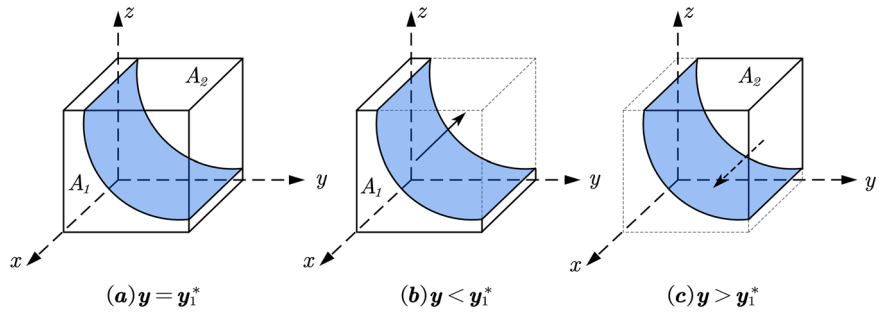

Fig. 8 | Phase diagram of strategy evolution for DMAs. Cubic phase diagrams depict stable strategy regions for DMAs at varying values of y relative to critical threshold y_1^*.

equilibrium in stabilizing data operations, and the differentiated efficacy of government interventions. By proposing tiered regulatory designs, risk-balanced revenue distribution systems, and technology-enhanced governance models, the research bridges theoretical innovation with actionable policy solutions. These contributions not only advance the scholarly discourse on data factor marketization but also provide a structured pathway for institutional reforms in China's healthcare data governance. Future studies could extend this framework to cross-border data flow scenarios or incorporate real-world transaction data to enhance model generalizability, further enriching the governance toolkit for the data-driven healthcare era.

As an exploratory study, this research offers novel insights into healthcare data governance using an evolutionary game-theoretic framework, but it also has several limitations. The model assumes bounded rationality among stakeholders, yet real-world decision-making in medical data sharing is influenced by multifaceted subjective factors such as trust dynamics, experiential biases, and organizational cultural influences. Additionally, the parameter calibration is largely based on China's institutional context, which may limit its direct applicability to regions with different regulatory philosophies. Lastly, the payoff functions prioritize transparency over computational complexity, which aligns with regulatory objectives but restricts direct comparability with advanced evolutionary game models in operations research.

Future studies could address these limitations by incorporating the cognitive, affective, and conative dimensions of game agents into system dynamics modeling, developing phased multi-stage repeated game frameworks that account for market maturation, and constructing more nuanced payoff functions that better reflect real-world stakeholder interactions.

Methods

Evolutionary game model construction and analysis

The application of evolutionary game theory has expanded significantly beyond its biological origins, establishing itself as a robust analytical framework across disciplines such as behavioral economics, management science, sociology, and environmental studies[49–52]. In recent years, this theoretical approach has gained traction in analyzing digital economy dynamics and data governance challenges. For instance, scholars have modeled interactions among open data managers to unravel social dilemmas inherent in decision-making processes, particularly emphasizing the role of cooperation in balancing public and private interests[53]. Further studies have quantified behavioral patterns through variables such as data acquisition costs, open data value, regulatory incentives, and users' analytical capabilities, offering foundational insights into data governance mechanisms[54].

Sector-specific applications of evolutionary game theory have proliferated, with notable contributions in manufacturing big data consortia[55], logistics platforms[56], and healthcare data ecosystems[57]. However, three critical limitations persist in current research. First, existing frameworks predominantly focus on dyadic interactions, overlooking the tripartite evolutionary dynamics involving governments, operational entities, and data subjects within principal-agent relationships. Second, healthcare-specific models frequently employ oversimplified parameter assumptions that fail to capture the institutional complexity of healthcare data markets, resulting in conclusions with limited practical relevance. Third, few studies systematically integrate the unique governance structures of emerging economies, particularly China's hybrid model blending administrative oversight with market-driven mechanisms.

This study addresses these research gaps by developing a replicable framework that integrates theoretical rigor with contextual specificity. By bridging evolutionary game theory with practical governance challenges in vertical domains, our model advances methodological applications while providing emerging economies with actionable strategies to balance security-efficiency trade-offs in healthcare data markets.

Strategic stability analysis of DMAs

From Fig. 2, the expected payoffs for the DMAs under strong and weak incentive strategies, denoted as π_{11} and π_{12}, as well as the corresponding average expected payoff $\overline{\pi_1}$, can be obtained.

$$\begin{aligned}
\pi_{11} &= yz(R_1 - S_1 - V_1 - F_1) + y(1-z)(R_1 - S_1 - F_1) \\
&\quad + (1-y)z[R_1 - V_1 + (1+\delta)n(P - I_1)] + (1-y)(1-z)[R_1 + n(P - I_1)] \\
\pi_{12} &= yz(R_2 - S_2 - V_2 - F_2) + y(1-z)(R_2 - S_2 - F_2) \\
&\quad + (1-y)z[R_2 - V_2 + n(P - I_1)] + (1-y)(1-z)[R_2 + n(P - I_1)] \\
\overline{\pi}_1 &= x\pi_{11} + (1-x)\pi_{12}
\end{aligned} \tag{2}$$

From this, the equation for the replication dynamics of the DMAs strategy selection can be obtained as:

$$\begin{aligned}
F(x) &= \tfrac{dx}{dt} = x(\pi_{11} - \bar{\pi}_1) = x(1-x)(\pi_{11} - \pi_{12}) \\
&= x(1-x)[(R_1 - R_2) + (S_2 - S_1 + F_2 - F_1)y + n(P - I_1)z - [\delta n(P - I_1)]yz]
\end{aligned} \tag{3}$$

The derivative of x is:

$$\begin{aligned}
\tfrac{dF(x)}{dx} &= (1-2x)[(R_1 - R_2) + (S_2 - S_1 + F_2 - F_1)y \\
&\quad + n(P - I_1)z - [\delta n(P - I_1)]yz]
\end{aligned} \tag{4}$$

For simplification, let:

$$G(y) = (R_1 - R_2) + (S_2 - S_1 + F_2 - F_1)y + n(P - I_1)z - [\delta n(P - I_1)]yz \tag{5}$$

According to the micro-dynamics stability criterion, the DMAs obtains the strong stable state that satisfies:$F(x) = 0$, and $\frac{dF(x)}{dx} < 0$.

From $G(y^*) = 0$, we can obtain:

$$y_1^* = \frac{(R_2 - R_1) - n(P - I_1)z}{(S_1 - S_2 + F_1 - F_2) - \delta n(P - I_1)z} \tag{6}$$

$$z_1^* = \frac{R_2 - R_1}{n(P - I_1)} \tag{7}$$

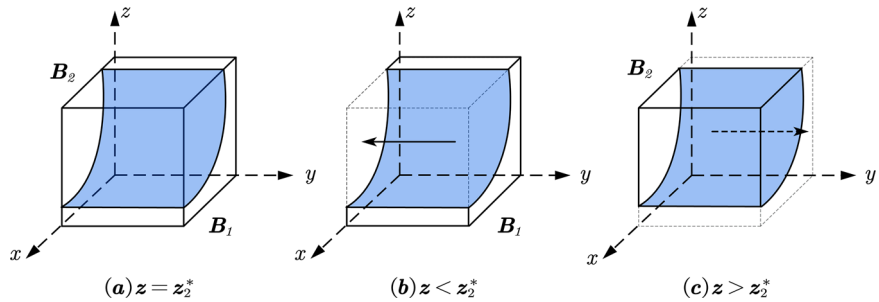

Fig. 9 | Phase diagram of strategy evolution for DODs. Cubic phase diagrams depict stable strategy regions for DODs at varying values of z relative to critical threshold z_2^*.

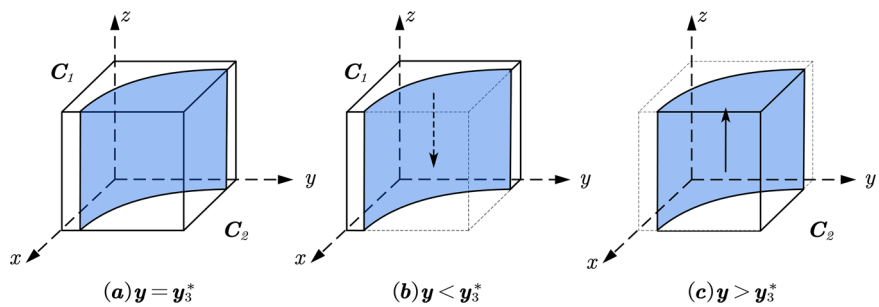

Fig. 10 | Phase diagram of strategy evolution for Des. Cubic phase diagrams depict stable strategy regions for DEs at varying values of y relative to critical threshold y_3^*.

Proposition 1. If $z > z_1^*$, then $x = 1$ is an evolutionarily stable strategy. If $z < z_1^*$, then: when $y = y_1^*$, all x are in a stable state; when $y < y_1^*$, $x = 1$ is an evolutionarily stable strategy; when $y > y_1^*$, $x = 0$ is the evolutionarily stable strategy at this time.

Proof. Since $\frac{\partial G(y)}{\partial y} < 0$, $G(y)$ is a decreasing function with respect to y. If $z > z_1^*$, then $y_1^* < 0$. For $y \in [0,\ 1]$, $G(z) > 0$, $F(1) = 0$, and $\frac{dF(y)}{dy}|_{x=1} < 0$, thus $x = 1$ is an evolutionarily stable strategy. If $z < z_1^*$, then: when $y = y_1^*$, $G(z) = 0$, $F(x) \equiv 0$, and $\frac{dF(x)}{dx} \equiv 0$, at this time all x are in a stable state; when $y < y_1^*$, $G(z) > 0$, $F(1) = 0$, and $\frac{dF(x)}{dx}|_{x=1} < 0$, so $x = 1$ is the evolutionarily stable strategy for the DMAs at this time; when $y > y_1^*$, $G(z) < 0$, $F(0) = 0$, and $\frac{dF(x)}{dx}|_{x=0} < 0$, thus $x = 0$ is the evolutionarily stable strategy at this time (Fig. 8).

Strategic stability analysis of DODs

From Fig. 2, the expected payoffs π_{21} and π_{22} under the compliance and profit-seeking strategies of DODs, as well as the corresponding average expected payoff $\overline{\pi_2}$, can be obtained.

$$\begin{aligned}
\pi_{21} &= xz[S_1 + (1+\delta)^2 mn(K - C_{high} - D) + F_1] + x(1-z)[S_1 + (1+\delta)mn(K - C_{high} - D) + F_1 - C] \\
&\quad + (1-x)z[S_2 + mn(K - C_{high} - D) + F_2] + (1-x)(1-z)[S_2 + mn(K - C_{high} - D) + F_2 - C] \\
\pi_{22} &= xz[(1+\delta)mn(K - C_{low} - D) - (1+\delta)nP] + x(1-z)[mn(K - C_{low} - D) - nP - C] \\
&\quad + (1-x)z[mn(K - C_{low} - D) - nP] + (1-x)(1-z)[mn(K - C_{low} - D) - nP - C] \\
\overline{\pi_2} &= y\pi_{21} + (1-y)\pi_{22}
\end{aligned} \tag{8}$$

From this, the equation for the replication dynamics of the DODs strategy selection can be obtained as:

$$\begin{aligned}
F(y) &= \tfrac{dy}{dt} = y(\pi_{21} - \overline{\pi_2}) = y(1-y)(\pi_{21} - \pi_{22}) \\
&= y(1-y)\left\langle \begin{matrix} [S_2 + mn(C_{low} - C_{high}) + F_2] + x[(S_1 - S_2) + \delta mn(K - C_{high} - D) + (F_1 - F_2)] \\ + xz\{(1+\delta)mn(K - D) + mn[(1+2\delta)C_{high} + \delta C_{low}] + \delta nP - C\} \end{matrix} \right\rangle
\end{aligned} \tag{9}$$

The derivative of y is:

$$\frac{dF(y)}{dy} = (1-2y)\left\langle \begin{matrix} [S_2 + mn(C_{low} - C_{high}) + F_2] + x[(S_1 - S_2) + \delta mn(K - C_{high} - D) + (F_1 - F_2)] \\ + xz\{(1+\delta)mn(K - D) + mn[(1+2\delta)C_{high} + \delta C_{low}] + \delta nP - C\} \end{matrix} \right\rangle \tag{10}$$

For simplification, let:

$$\begin{aligned}
J(z) &= [S_2 + mn(C_{low} - C_{high}) + F_2] + x[(S_1 - S_2) + \delta mn(K - C_{high} - D) + (F_1 - F_2)] \\
&\quad + xz\{(1+\delta)mn(K - D) + mn[(1+2\delta)C_{high} + \delta C_{low}] + \delta nP - C\}
\end{aligned} \tag{11}$$

According to the stability theorem of differential equations, for the DODs to maintain a stable probability of adhering to standards, it is necessary that: $F(y) = 0$, and $\frac{dF(y)}{dy} < 0$.

From $J(z^*) = 0$, we can obtain:

$$z_2^* = \frac{-[S_2 + mn(C_{low} - C_{high}) + F_2] - x[(S_1 - S_2) + \delta mn(K - C_{high} - D) + (F_1 - F_2)]}{\{(1+\delta)mn(K - D) + mn[(1+2\delta)C_{high} + \delta C_{low}] + \delta nP - C\}x} \tag{12}$$

$$x_2^* = \frac{[S_2 + mn(C_{low} - C_{high}) + F_2]}{-[(S_1 - S_2) + \delta mn(K - C_{high} - D) + (F_1 - F_2)]} \tag{13}$$

Table 4 | Simulation parameter assignment methods and rationale

Parameters	Initial value (10,000 Yuan)	Calculation method and basis	Data source
Policy Incentive Parameters			
R_1	13975.2	R_1 = GDP × Healthcare Expenditure Ratio × Benefit Multiplier = 129.4 trillion × 7.2% × 15%	① 2023 China GDP total 12.94 trillion (National Bureau of Statistics) [a]; ② Healthcare total expenditure ratio (7.2%) (China Health Statistics Yearbook); ③ Strong incentive data circulation promotion rate (15%) [b]
R_2	4658.4	R_2 = GDP × Healthcare Expenditure Ratio × Benefit Multiplier = 129.4 trillion × 7.2% × 5%	① Weak incentive data circulation promotion rate (5%)
F_1	1418.74	F_1 = (2675billion ÷ 12032 ÷ 8%) × 4% × (1 + 5%)^5	① 2018 public hospital financial subsidy data [c]; ② National Health Information Development Plan (2003-2010) suggests information technology investment standard upper limit (4%) [d]; ③ 5% discount rate compound interest calculation [e]
F_2	709.37	F_2 = (2675billion ÷ 12032 ÷ 8%) × 2%×(1 + 5%)^5	① National Health Information Development Plan (2003-2010) suggests information technology investment standard lower limit (2%)
S_1	100	Take the upper limit value of Dalian DCMM certification reward standard (1 million yuan)	① Dalian supports the high-quality development of software and information technology service industry policies [f]
S_2	10	Take the lower limit value of Shenzhen DCMM certification reward (100,000 yuan)	① Bao'an District measures to promote the high-quality development of software and internet industry [g]
V_1	0.01	Hangzhou traffic data sharing points exchange mechanism (average daily points exchange value)	① 2024 Union Pay Business City Data Sharing Report
V_2	0	No relevant policy support	——
Data Governance Parameters			
C_{high}	246.375	C_{high} = 821.25TB × 3000 yuan/TB	① Average annual data volume of enterprises: 32.85 ZB ÷ 40 million enterprises = 821.25 TB per enterprise (National Data Resource Survey Report (2023)) [h]; ② Data governance costs of a Chinese big data company (3,000 yuan/TB) [i]
C_{low}	492.75	C_{low} = C_{high} × 2	① IDC industry benchmark (low-quality data governance cost is twice that of high quality)
P	50	Penalty range	① Article 45 of the People's Republic of China Data Security Law [j]
D	7130.73	D = 9.77 million USD × 7.2983	① Healthcare industry leakage cost is 200% of the industry average (Ponemon Institute 2024 study) [k]
K	3542.86	K = 248 million yuan ÷ 13	① 2023 Audit Office annual report
C	1798.19	C = (C_{high} + C_{low}) ÷ 2 × 20%	① Industry research assumption, average data governance cost × additional cost coefficient
Risk and Externality Parameters			
I_1	2189.49	I_1 = 3million USD × 7.2983	① Average reputation loss of data breaches for medium-sized enterprises in the United States (IBM Data Breach Cost Report); ② Central Bank exchange rate on December 20, 2024
I_2	0.09	Industry data estimation	① Sage Consulting tracking study on the implementation rate of AI strategy in medical institutions [l]
λ	0.029	Effective data utilization rate = 2.9% × 60%	① 40% of the national stored data is idle (National Data Resource Survey Report (2023)) [m]
ΔU	526000	2022 China medical data transaction scale 5.26 billion yuan	① 2023 Shanghai Data Exchange China Data Trading Market Research and Analysis Report
H	255.19	H = (330 million+63 million) ÷ 154	① Public data from Xianyang City Health Commission, Shaanxi Province [n]
Structural Parameters			
m	4	Assigned by equation balance principle	① 2023 Audit Office report on the central budget execution and other financial revenues and expenditures [o]
n	13	Assigned by equation balance principle	① 2023 Audit Office annual report; ② State Council organizational setup documents [p]

Table 4 (continued) | Simulation parameter assignment methods and rationale

Parameters	Initial value (10,000 Yuan)	Calculation method and basis	Data source
N	26	Assigned by equation balance principle	① 2023 Audit Office annual report; ② State Council organizational setup documents
δ	2	Strong incentive policy effectiveness is twice that of weak incentives	① Comparative analysis of policy effects between Dalian and Daxing

[a]On December 26, 2024, the National Bureau of Statistics revised the GDP for the year 2023 to 129.4 trillion yuan, an increase of 3.4 trillion yuan from the preliminary figure, representing a 2.7% growth.
[b]Du Wei, Director of the Digital Technology and Infrastructure Construction Department of the National Data Bureau, stated, "According to estimates, a 10% increase in data flow will drive a 0.2% increase in GDP, with data circulation averaging a 10% boost to industry profits."
[c]In 2018, the total financial subsidy for public hospitals in China amounted to 267.5 billion yuan (across 12,032 hospitals), accounting for 8–10% of their total revenue.
[d]The "National Health Information Development Plan (2003-2010)" suggested that hospitals should allocate 2–4% of their total revenue to information technology construction. The Sichuan Health Commission issued the "Digital Hospital Evaluation Standards," requiring hospitals at all levels to budget no less than 2% of their business income specifically for information technology construction to ensure the development of smart hospitals.
[e]Reference was made to the central bank's medium-term lending facility interest rate.
[f]Enterprises that pass the CMMI, CSMM, ITSS, and DCMM certifications for the first time are awarded 100,000 to 1 million yuan based on their certification level.
[g]Enterprises that pass the DCMM Level 2 or above certification in the software industry are granted a one-time reward of 100,000 yuan.
[h]A report by the National Data Resource Survey Working Group indicated that in 2023, the total data production in China reached 32.85 zettabytes (ZB). According to 2023 statistics, there are approximately 40 million enterprises in China, including private, state-owned, and foreign-funded enterprises. Data production is unevenly distributed across industries, with technology-intensive sectors like the internet, e-commerce, finance, and manufacturing generating substantial amounts of data, while traditional manufacturing and agriculture produce less. Data generation in the healthcare sector can be assumed to be at the national average level.
[i]Due to the lack of specific cost data for healthcare data governance, we reference the unit cost from Xiaomi Technology, which ranks ninth among the "Top 50 Big Data Companies in China 2023." Xiaomi Technology's annual data management cost for 1000 terabytes (TB) of data is 3 million yuan. This approach ensures a conservative estimate of C_{high} in our simulation.
[j]Organizations or individuals conducting data processing activities who fail to fulfill the data security protection obligations stipulated in Articles 27, 29, and 30 of this law shall be ordered by the competent authorities to correct their actions, warned, and may be fined between 50,000 and 500,000 yuan. Directly responsible supervisors and other directly liable personnel may be fined between 10,000 and 100,000 yuan. Refusal to correct or causing serious consequences such as significant data leaks may result in fines between 500,000 and 2 million yuan, and may also lead to the suspension of related business, business rectification orders, revocation of related business licenses, or business licenses, with fines for directly responsible supervisors and other directly liable personnel ranging from 50,000 to 200,000 yuan.
[k]Between March 2023 and February 2024, the average global cost of data breaches rose to 4.88 million USD. In various industries, the healthcare sector leads with the highest average cost per breach at 9.77 million USD.
[l]A report released by Sage Growth Partners, a healthcare consulting firm, in 2021 showed that 90% of hospitals in the United States have formulated AI/automation strategies, but only 9% have fully implemented their strategies.
[m]The "National Data Resource Survey Report (2023)" compiled by the National Data Resource Survey Working Group indicates that in 2023, China's total data production reached 32.85 ZB, with only 2.9% of the data being preserved. Among the currently stored data, approximately 40% has not been used for a year.
[n]Xianyang City in Shaanxi Province has established health records for 4.92 million residents by uploading basic resident information, medical treatment, and health status to the blockchain. Through the supervision and screening of medical data, it has reduced excessive prescriptions by approximately 29% and overmedical treatment by 21%, alleviating the burden on the public by 330 million yuan; it has also handled 275 cases of prescription fraud, saving more than 63 million yuan in medical insurance funds for the government. As of 2018, Xianyang City had a total of 154 hospitals.
[o]The "Audit Report on the Execution of the Central Budget and Other Financial Revenues and Expenditures in 2023" by the National Audit Office revealed that seven maintenance units under four departments, without approval, self-determined data content, service forms, and charging standards, and illegally charged external parties based on data from 13 systems, amounting to 248 million yuan.
[p]The State Council of the People's Republic of China comprises 26 constituent departments.

Proposition 2. If $x > x_2^*$, then $y = 1$ is always an evolutionarily stable strategy; if $x < x_2^*$, then: when $z = z_2^*$, all y are in a stable state; when $z < z_2^*$, $y = 0$ is the evolutionarily stable strategy; when $z > z_2^*$, $y = 1$ is the evolutionarily stable strategy.

Proof. Since $\frac{\partial J(z)}{\partial z} > 0$, $J(z)$ is an increasing function with respect to z. If $x > x_2^*$, then $z_2^* < 0$, and for $z \in [0,\ 1]$, $J(z) > 0$, $F(1) = 0$, and $\frac{dF(y)}{dy}|_{y=1} < 0$, thus $y = 1$ is an evolutionarily stable strategy. If $x < x_2^*$, then: when $z = z_2^*$, $J(z) = 0, F(y) \equiv 0$, and $\frac{dF(y)}{dy} \equiv 0$, at this time all y are in a stable state; when $z < z_2^*$, $J(z) < 0$, $F(0) = 0$, and $\frac{dF(y)}{dy}|_{y=0} < 0$, so $y = 0$ is the evolutionarily stable strategy; when $z > z_2^*$, $J(z) > 0$, $F(1) = 0$, and $\frac{dF(y)}{dy}|_{y=1} < 0$, thus $y = 1$ is the evolutionarily stable strategy at this time (Fig. 9).

Strategic stability analysis of DEs

From Fig. 2, the expected payoffs π_{31} and π_{32} under the authorization and non-authorization strategies of DEs, as well as the corresponding average expected payoff $\overline{\pi_3}$, can be obtained.

$$\begin{aligned}\pi_{31} &= xy[V_1 + (1+\delta)nI_2 + \tfrac{(1+\delta)^2 mn}{N}H + \lambda\Delta U] + x(1-y)[V_1 + \tfrac{(1+\delta)mn}{N}H + \Delta U] \\ &\quad + (1-x)y[V_2 + nI_2 + \tfrac{mn}{N}H + \lambda\Delta U] + (1-x)(1-y)[V_2 + \tfrac{mn}{N}H + \Delta U] \\ \pi_{32} &= xy[nI_2 + \tfrac{(1+\delta)mn}{N}H] + x(1-y)(\tfrac{mn}{N}H) \\ &\quad + (1-x)y(nI_2 + \tfrac{mn}{N}H) + (1-x)(1-y)(\tfrac{mn}{N}H) \\ \overline{\pi}_3 &= z\pi_{31} + (1-z)\pi_{32}\end{aligned} \tag{14}$$

From this, the equation for the replication dynamics of the DEs strategy selection can be obtained as:

$$\begin{aligned}F(z) &= \tfrac{dz}{dt} = z(\pi_{31} - \overline{\pi}_3) = z(1-z)(\pi_{31} - \pi_{32}) \\ &= z(1-z)[V_2 + \Delta U + x[(V_1 - V_2) + \tfrac{\delta mn}{N}H] + y(\lambda - 1)\Delta U + xy(\delta nI_2 + \tfrac{\delta^2 mn}{N}H)]\end{aligned} \tag{15}$$

The derivative of z is:

$$\frac{dF(z)}{dz} = (1-2z)\left[V_2 + \Delta U + x\left[(V_1 - V_2) + \frac{\delta mn}{N}H\right] + y(\lambda - 1)\Delta U + xy\left(\delta nI_2 + \frac{\delta^2 mn}{N}H\right)\right] \tag{16}$$

For simplification, let:

$$K(y) = V_2 + \Delta U + x\left[(V_1 - V_2) + \frac{\delta mn}{N}H\right] + y(\lambda - 1)\Delta U + xy\left(\delta nI_2 + \frac{\delta^2 mn}{N}H\right) \tag{17}$$

According to the stability theorem of differential equations, for the DEs to maintain a stable probability of granting authorization, it is necessary that: $F(z) = 0$, and $\frac{dF(z)}{dz} < 0$.

From $K(y_3^*) = 0$, we can obtain:

$$y_3^* = \frac{V_2 + \Delta U + x[(V_1 - V_2) + \frac{\delta mn}{N} H]}{-(\lambda - 1)\Delta U - x(\delta n I_2 + \frac{\delta^2 mn}{N} H)} \quad (18)$$

$$x_3^* = \frac{-(V_2 + \Delta U)}{(V_1 - V_2) + \frac{\delta mn}{N} H} \quad (19)$$

Proposition 3. If $x > x_3^*$, then $z = 1$ is always an evolutionarily stable strategy; if $x < x_3^*$, then: when $y = y_3^*$, all z are in a stable state; when $y < y_3^*$, $z = 0$ is the evolutionarily stable strategy; when $y > y_3^*$, $z = 1$ is the evolutionarily stable strategy.

Proof. Since $\frac{\partial K(y)}{\partial y} > 0$, $K(y)$ is an increasing function with respect to y. If $x > x_3^*$, then $y_3^* < 0$, and for $y \in [0, 1]$, $K(y) > 0$, $F(1) = 0$, and $\frac{dF(z)}{dz}|_{z=1} < 0$, thus $z = 1$ is an evolutionarily stable strategy. If $x < x_3^*$, then: when $y = y_3^*$, $K(y) = 0$, $F(z) \equiv 0$, and $\frac{dF(z)}{dz} \equiv 0$, at this time all z are in a stable state; when $y < y_3^*$, $K(y) < 0$, $F(0) = 0$, and $\frac{dF(z)}{dz}|_{z=0} < 0$, so $z = 0$ is the evolutionarily stable strategy; when $y > y_3^*$, $K(y) > 0$, $F(1) = 0$, and $\frac{dF(z)}{dz}|_{z=1} < 0$, thus $z = 1$ is the evolutionarily stable (Fig. 10).

Simulation analysis

While the Jacobian matrix can assess the stability of equilibrium points in evolutionary systems, it falls short in delineating the realization process of equilibria or the impact of parameters on evolutionary trajectories. To intuitively understand the evolutionary stable strategies (ESS) of DMAs, DODs, and DEs under varying conditions, this study employs numerical simulations using Python 3.9.13 (Python Software Foundation, Delaware, USA) to investigate how key variables influence stakeholders' strategic behaviors and drive the system toward desirable stable states. Notably, the ESS configurations $E_7(0, 1, 1)$ (Weak Incentives, Compliance, Authorization) and $E_8(1, 1, 1)$ (Strong Incentives, Compliance, Authorization) represent the expected regulatory outcomes under distinct incentive regimes.

Parameter assignment

To ensure simulation validity, parameters are assigned based on China's healthcare data circulation context, incorporating literature reviews, expert interviews, industry surveys, and equilibrium principles. The methodology is structured as follows: (1) The policy incentive parameters are constructed using the dynamic benchmark method. The basic public revenues R_1 and R_2 are set with the 2023 GDP as the benchmark value, in conjunction with the data flow promotion rate stratified by the National Data Bureau. The financial subsidies F_1 and F_2 are derived from historical data on financial subsidies for public hospitals, with their present value calculated through a discounting model. The operating sector incentives S_1 and S_2, as well as the individual incentives V_1 and V_2, are directly referenced from provincial and municipal policy documents, reflecting the reward intervals for extreme values. (2) Data governance parameters are measured using a unit cost model. The governance costs C_{high} and C_{low} are determined based on the ratio of total national data to the number of enterprises, which defines the unit data volume. These are further calibrated with industry practices and benchmark unit prices from the IDC. The violation revenue K and penalties P are based on audit report disclosure cases and the statutory fines outlined in the 'Data Security Act', specifically the lower-end assignments. The additional operating costs C are based on the average governance cost, with a 20% industry-specific correction factor applied. (3) Risk and externality parameters are introduced through international benchmarking and domain-specific correction. The moral risk loss I_1 and data leakage loss D are quoted from IBM and Ponemon report data, with a 200% correction factor applied, reflecting the characteristics of the healthcare industry. Trust I_2 and positive externalities ΔU and λ are empirically assigned using industry research data and transaction scale. Health benefits H are inferred through an inversion method, utilizing Xianyang City's healthcare data application case to deduce the contribution value of a unit organization. (4) The structural parameters adhere to the system dynamics equilibrium principle. The number of operating entities m, n, and departmental level N are determined based on audit reports and institutional setup documents from the State Council. The incentive factor δ is assigned using the expert assessment method, following a comparative analysis of policy effects. Based on the actual situation outlined above, the initial assignment for each parameter is derived from the sources and data shown in Table 4.

Data availability

All data and materials relevant to the study are included in the article or uploaded as supplementary information.

Code availability

All codes relevant to the study are accessible in the following GitHub repository [https://github.com/hannalyao/Evolutionary-Game-Theory-in-Healthcare-Data-Governance].

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Author contributions

H.Y. conducted the analysis and prepared the initial draft. Y.L. and H.Y. contributed to the conceptualization and investigation of the study, and reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Ethics

This work did not require ethical approval from a human subject or animal welfare committee.

Additional information

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Correspondence and requests for materials should be addressed to Yuanli Liu.

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